ELEC3104: Digital Signal Processing

Topic 1: DSP fundamentals

Model Answers

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

Topic 1: Pass-Level Tutorial Questions

1. An analogue signal contains frequencies up to 10 kHz. Suppose that we sample this signal at a sampling frequency f_s = 8kHz. What happens to the frequency f_1 = 5 kHz? (Tick the correct answer)
   a) 5kHz frequency will remain unchanged
   b) 5kHz will alias to 3kHz
   c) 5kHz will alias to 0Hz
   d) None of the above

f_1 (5 kHz) $> \frac{f_s}{2}$ (4 kHz) ∴ **Aliasing will occur.**
The new frequency is: $f_1' = f_1 - kf_s$
For $k = 1$, $f_1' = 5000 - 1(8000) = -3000$ Hz

2. Which of the following is not a periodic signal ? (Tick the correct answer)
   a) x[n] = sin[πn/9] — Period = 18
   b) x[n] = sin[nπ] — Period = 2
   c) x[n] = cos[πn²/$\sqrt{15}$] — Non-periodic (because $\sqrt{15}$ is irrational).
   d) x[n] = sin[πn/5 + π] + cos[πn/10 – π] — Period = Least Common Multiple of $N1$ and $N2$ $= \mathrm{LCM}(10, 20) = 20$
      Period N1 = 10; Period N2 = 20

3. Given a real-valued bandpass signal x_a(t) whose spectrum of X_a(f) nonzero only for $9\ kHz < |f| < 12kHz$, determine the minimum sampling frequency , f_s , for which no aliasing occurs? (Tick the correct answer)
   a) f_S = 18 kHz
   b) f_S = 6 kHz
   c) f_S = 24 kHz
   d) None of the above

Bandwidth $(B) = 12 - 9 = 3$ kHz
$f_2 = 12$ kHz, which is an integer multiple of the bandwidth.
Therefore, $f_s(\text{minimum}) = 2B = 2 \times 3$ kHz = = 6 kHz

4. A discrete-time signal $x[n]$ = 6.35 cos(nπ/10) is quantised with a resolution Δv = 0.2 $volts$. Assuming ADC covers the full signal range , how many bits are required? (Tick the correct answer)
   a) 6 bits
   b) 8 bits
   c) 12 bits

$\Delta v = \frac{2 \times 6.35}{2^B}$; $\Delta v = 0.2$ (given)

Therefore, $2^B = \frac{2 \times 6.35}{0.2} = 63.5$; Hence, $\boxed{B = 6 \text{ bits}}$

Topic 1: Pass-Level Tutorial Questions

5. Consider the amplitude modulated signal $x(t) = 5\,cos(3\pi t)\,cos(\pi t)$. Determine the frequency components [you may assume that the frequencies are in kHz) (Tick the correct answer)
   a) $\pm$1.5kHz and $\pm$0.5kHz
   b) $\pm 3\text{kHz}$ and $\pm$1kHz
   c) $\pm$2kHz
   d) $\pm$2kHz and $\pm$1kHz

$$x(t) = 5\cos(2 \times 1.5\pi t) \cdot \cos(2 \times 0.5\pi t)$$

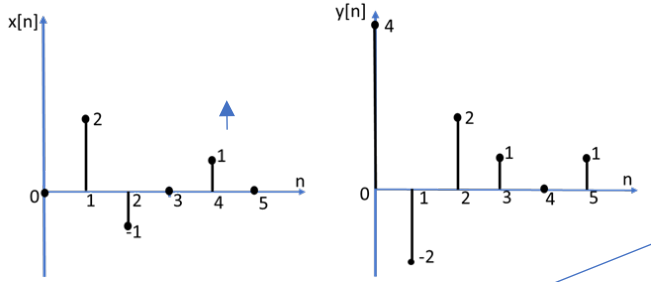
Where $f_1 = 1.5$ kHz, $f_2 = 0.5$ kHz

The product of two real sinusoids results in the **sum and difference frequencies**: $\pm(f_1 + f_2)$ and $\pm(f_1 - f_2)$

Therefore,

± 2 kHz and $\pm$ 1 kHz

6. Given the discrete-time sequences $x[n]$ and $y[n]$ shown, which difference equation best describes their relationship ? (Tick the correct answer)



$x[0] = 0;\ y[0] \neq 0\ (y[0] = 4)$

Therefore, the signal $x[n]$must have been **advanced by one sample**.

   a) y[n] = x[n-1] + 2x[n+1]
   b) y[n] = 2x[n] – x[n-1]
   c) y[n] = 2x[n-1] + x[n-2]
   d) y[n] = 2x[n-2] – x[n]

7. Determine the period of the discrete time signal $x[n] = \sin(0.2n + \pi)$. (Tick the correct answer)

   (a) 1 sample (b) 5 samples
   (c) 10 π samples (d) Non of the above

The signal is not periodic. $\omega_o = 0.2$ rad/sample

$\omega_o/(2\pi) = 1/(10\pi)$, which is irrational.

See lecture notes Example 4

Topic 1: Pass-Level Tutorial Questions

1. A sinusoidal signal is sampled at 50 kHz using a uniform B-bit A/D converter. Assuming that the sinusoid uses the full input range of the A/D converter, determine the minimum number of bits required to achieve an SQNR of at least 92 dB.

Minimum number of bits for SQNR ≥ 92 dB

For a full-scale sinusoidal input to a uniform B-bit A/D converter:

SQNR ≈ 6.02B + 1.76 dB; 6.02B + 1.76 ≥ 92

B ≥ (92 - 1.76)/6.02 = 14.99

Therefore, the minimum number of bits is B = 15 bits.

2. Show that the signal to quantisation noise ratio (SQNR) of a linear B-bit analogue-to-digital converter is given by $SQNR = 6.02B + 4.77 - 20\log\left({}^{A}/_{\sigma_{sig}}\right)$ where the input range of the A/D converter is $\pm$A and the rms value of the input signal is σ_{sig}. Determine the SQNR if *B* is 16 bits and the input is a signal with a rms value of A/5.

$\text{SQNR} = 10\log_{10}\left(\frac{\sigma_S^2}{\sigma_e^2}\right)$; Since $\sigma_e^2 = \frac{(\Delta v)^2}{12}$ and $\Delta v = \frac{2A}{2^B}$, then $\text{SQNR} = 10\log_{10}\left[\frac{\sigma_S^2}{\left(\frac{2A}{2^B}\right)^2/12}\right] = 10\log_{10}\left(\frac{\sigma_S^2\ 2^{2B}\times 3}{A^2}\right)$

Therefore, $\text{SQNR} = 10\log_{10}\left(\frac{\sigma_S^2}{A^2}\right) + 10\log_{10}(2^{2B}) + 10\log_{10}(3) = 20\log_{10}\left(\frac{\sigma_S}{A}\right) + B(10\log_{10}4) + 4.77$

Since

$$\boxed{\text{SQNR} = 6.02B + 4.77 - 20\log_{10}\left(\frac{A}{\sigma_S}\right)}$$

For $B = 16$ and $\sigma_s = \sigma_{\text{sig}} = \frac{A}{5}$, $\text{SQNR} = 6.02(16) + 4.77 - 20\log_{10}\left(\frac{A}{\frac{A}{5}}\right) = 6.02(16) + 4.77 - 20\log_{10}(5)$

$$\boxed{\text{SQNR} \approx 87.1\ \text{dB}}$$

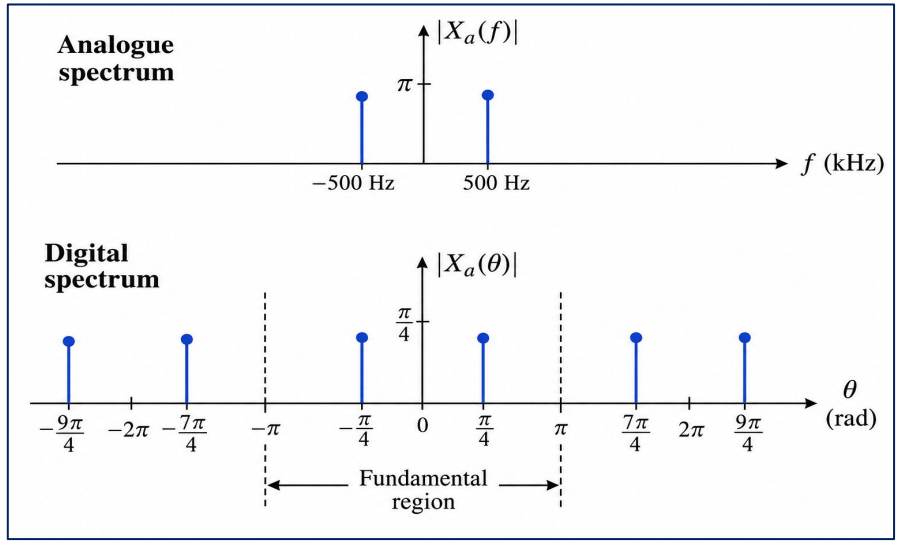
Topic 1: Pass-Level Tutorial Questions

3. An analogue signal $x_a(t) = \cos(2\pi 500t)$ is sampled at a rate of f_s = 4kHz. Determine the resulting analogue and digital magnitude spectra.

$$\mathcal{F}\mathrm{T}\{\cos(\omega_0 t)\} = \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$

Given $x_a(t) = \cos(2\pi 500t)$, therefore, $f_1 = 500$ Hz

The corresponding digital frequency is $\theta = 2\pi\frac{f_1}{f_s} = \frac{2\pi\times 500}{4000} = \frac{\pi}{4}$



4. An analogue signal $x(t)$ = sin(480πt) + 3sin(720πt) is sampled 600 times per sec:

(a) Determine the Nyquist sampling rate for $x(t)$.

$$x(t) = \sin(2\pi \times 240t) + 3\sin(2\pi \times 360t); \quad f_1 = 240 \text{ Hz}, \quad f_2 = 360 \text{ Hz}$$

Therefore, $f_{\max} = 360$ Hz and the Nyquist sampling rate is $f_{\text{Nyquist}} = 2f_{\max} = 720$ Hz

(b) Determine the folding frequency (or half the sampling frequency).

$f_s = 600$ Hz Therefore, $f_{\text{fold}} = \frac{f_s}{2} = 300$Hz

(c) What are the frequencies, in radians, in the resulting discrete time signal $x[n]$?

$f_2 > \frac{f_s}{2}$; Therefore, f_2 is affected by aliasing.

The new frequency due to aliasing is $f_2' = f_2 - kf_s$

For $k = 1$, $f_2' = 360 - 1(600) = -240\ Hz$

Therefore, $x[n] = \sin\left(2\pi\frac{240}{600}n\right) + 3\sin\left(2\pi\frac{-240}{600}n\right)$

Hence, $x[n] = -2\sin\left(2\pi\frac{240}{600}n\right) = -2\sin\left(\frac{4\pi}{5}n\right)$

(d) If $x[n]$ is passed through an ideal D/A converter what is the reconstructed signal $y(t)$?

$x[n] = -2\sin\left(2\pi\frac{240}{600}n\right)$; We have $t = nT = \frac{n}{f_s} = \frac{n}{600}$

Therefore, $y(t) = -2\sin(2\pi \times 240t)$

Hence, $y(t) = -2\sin(480\pi t)$

Topic 1: Pass-Level Tutorial Questions

5. A signal of $v(t) = 4.5\sin(1000\pi t)$ Volts is sampled at 2.5 kHz. An eight-bit A/D converter with a nominal input range of ± 5 Volts is used to convert sampled voltages to integer codes.
   (a) Calculate the quantization interval of the A/D converter.
   (b) Determine the maximum quantization error.

(a) Peak-to-peak amplitude: $V_{pp} = 10$ V

Quantisation interval (or resolution): $\Delta V = \frac{10}{2^B}$

For an 8-bit A/D converter, $\Delta V = \frac{10}{2^8}$; $\boxed{\Delta V = 39.06\text{mV}}$

(b) Maximum quantisation error: $e_{max} = \frac{\Delta V}{2}$; $\boxed{e_{max} = 19.53\text{ mV}}$

6. Consider the discrete-time sequence x[n] = cos[(nπ/8) + (π/4)] .
   (i) Determine the fundamental period of $x[n]$.
   (ii) If the sampling frequency f_s = 10 kHz, determine one corresponding analogue signal $x(t)$, assuming its frequency is below the Nyquist frequency.

(i) $x[n] = \cos\left(n\frac{\pi}{8} + \frac{\pi}{4}\right)$; Digital frequency: $\theta = \frac{\pi}{8}$

The fundamental period is $N = \frac{2\pi k}{\theta}$

For $k = 1$, $N = \frac{2\pi}{\pi/8} = 16$ samples . Therefore, $\boxed{\text{Period} = 16 \text{ samples}}$

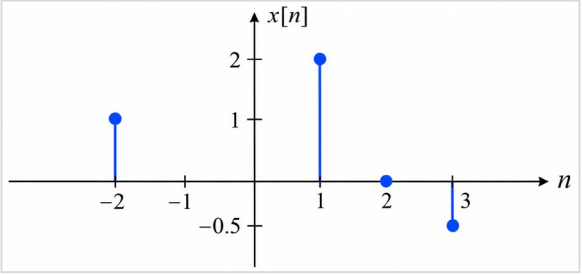
(ii) $\theta = \frac{\pi}{8}$; Since $\theta = 2\pi\frac{f_a}{f_s}$, $2\pi\frac{f_a}{f_s} = \frac{\pi}{8}$; Therefore, $\frac{f_a}{f_s} = \frac{1}{16}$

With $f_s = 10\,000$ Hz, $f_a = \frac{10\,000}{16} = 625$ Hz

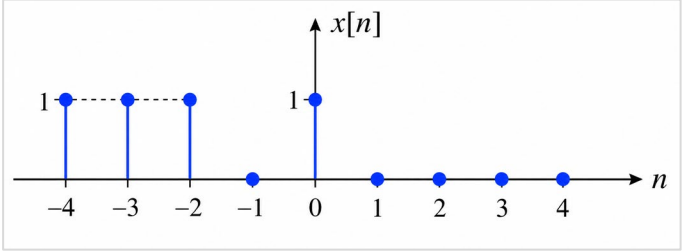
Hence, $\boxed{x(t) = \cos\left(2\pi(625)t + \frac{\pi}{4}\right)}$

7. Sketch the following:

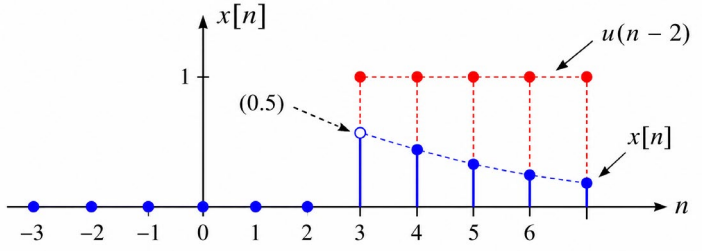
(a) x[n] = δ[n+2] + 2 δ[n-1] - 0.5 δ[n-3]



(b) x[n] = u[-n-2]

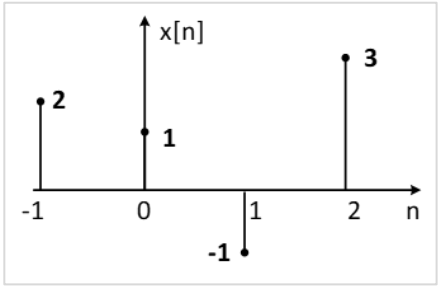


(c) x[n] = 0.5^n u[n-2]



Topic 1: Pass-Level Tutorial Questions

8. Write the equation for x[n] (see diagram) as a sum of scaled and delayed unit impulses



$$\boxed{x[n] = 2\delta(n+1) + \delta(n) - \delta(n-1) + 3\delta(n-2)}$$

9. For the system shown below, the input x(t) = cos(400πt) + cos(600πt). Find the output y[n] where $H(\theta) = \begin{cases} 1 & |\theta| \le 0.5\pi \\ 0 & otherwise \end{cases}$

$T = 0.001$ s; Therefore, the sampling frequency is $f_s = \frac{1}{T} = \frac{1}{0.001} = 1000$ Hz

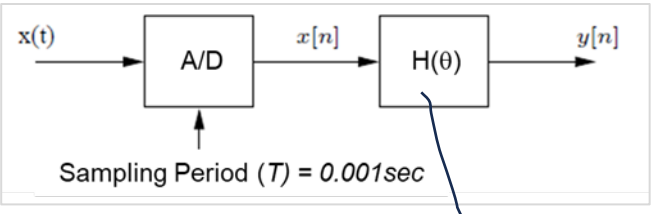
Hence, sampling $x(t)$gives the discrete-time signal $x[n]$:

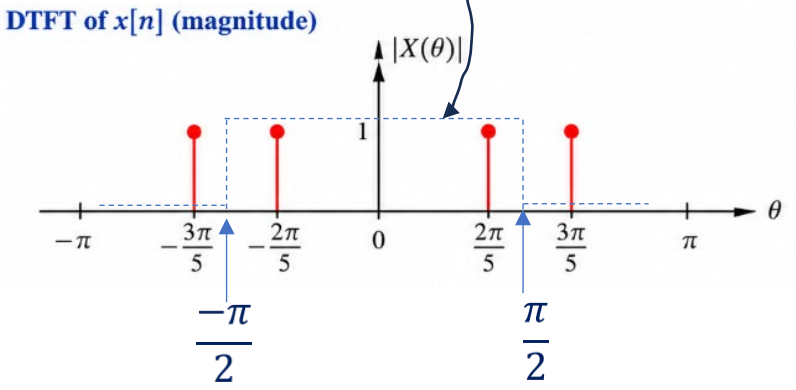
$$x[n] = \cos\left(2\pi\frac{200}{1000}n\right) + \cos\left(2\pi\frac{300}{1000}n\right)$$

$$\boxed{x[n] = \cos\left(\frac{2\pi}{5}n\right) + \cos\left(\frac{3\pi}{5}n\right)}$$

The filter bandwidth is $\frac{\pi}{2} = 0.5\pi$; Since $\frac{3\pi}{5} > \frac{\pi}{2}$, the frequency components at $\pm\frac{3\pi}{5}$ are removed by the filter $H(\theta)$, while the components at $\pm\frac{2\pi}{5}$ remain.

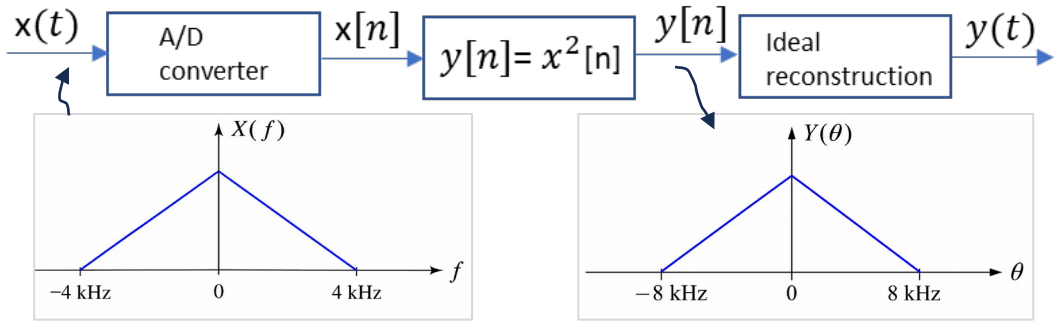
Therefore, $\boxed{y[n] = \cos\left(\frac{2\pi}{5}n\right)}$





Topic 1: Pass-Level Tutorial Questions

10. A speech signal is band limited to 4 kHz is ideally sampled at a sampling rate, f_s. the resulting discrete-time signal $x[n]$, is squared to give $y[n] = x^2[n]$, and $y[n]$ is ideally reconstructed to produce $y(t)$. What is the minimum sampling frequency f_s, required to ensure $y(t) = x^2(t)$?



Let $x(t) = A\sin(\omega_0 t)$;

$$y(t) = [x(t)]^2 = A^2\sin^2(\omega_0 t) = \frac{A^2}{2}(1 - \cos 2\,\omega_0 t)$$

(Squaring operation produces $2\omega_0$= 8 kHz)

Therefore, the spectrum of $y(t)$extends to twice the frequency of $x(t)$.

∴ Minimum sampling frequency should be 16 kHz

11. The signal $x(t) = 2\cos(2000\pi t) - 4\sin(4000\pi t)$ is quantised by rounding, using a 12-bit quantiser with an input range of ±6 V. What is the rms quantisation error and the SQNR?

The powers of the two sinusoidal components add.

The signal Power $P_s = \left(\frac{2}{\sqrt{2}}\right)^2 + \left(\frac{4}{\sqrt{2}}\right)^2 = 10$

Therefore, the RMS value of the signal is $x_{\text{rms}} = \sqrt{P_s} = \sqrt{10}$

The peak value of $x(t)$cannot exceed 6 V,

so the peak-to-peak amplitude is $V_{\text{pp}} = 12$ V

For a 12-bit quantiser, $\Delta V = \frac{12}{2^{12}} = \frac{12}{2^{12}} = 0.00293$ V

The RMS quantisation error is $\sigma_q = \frac{\Delta V}{\sqrt{12}} = \frac{12/2^{12}}{\sqrt{12}} = 0.0008457$ V

The SQNR in dB is $\text{SQNR} = 10\log_{10}\left(\frac{P_s\,(Signal\ Power)}{P_n(Noise\ Power)}\right)$;

where $P_n = \frac{(\Delta V)^2}{12}$

Therefore, $\text{SQNR} = 10\log_{10}\left[\frac{10}{(12/2^{12})^2/12}\right] = 71.46$ dB

Topic 1: Pass-Level Tutorial Questions

12. Consider a continuous time signal, $s(t)$, with a magnitude spectrum satisfying the following equation

$$|S(f)| = \begin{cases} 3 - |f| & 2\text{kHz} < |f| < 3\text{kHz} \\ 0 & \text{otherwise} \end{cases}$$

f – frequency in kHz. The signal $s(t)$ is sampled uniformly with a sampling frequency of 2kHz. Sketch the magnitude spectrum $|S(\theta)|$ of the resulting discrete –time signal over $-\pi \leq \theta \leq \pi$

The new frequency due to aliasing is: $f_o' = f_o - kf_s,\ k = 0, \pm1, \pm2, \ldots$

f_0 is the frequency present in the original analogue spectrum. The sampling frequency f_s = 2 kHz.

$f_o' = f_o - kf_s$

For $k = 1$:

$f_o' = 2 - 1(2) = 0$

$f_o' = 3 - 1(2) = 1$

For $k = -1$:

$f_o' = -2 + 1(2) = 0$

$f_o' = -3 + 1(2) = -1$

